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COURSE NAME – Operation System
COURSE CODE – CSA0408

Lab Observations

EXPERIMENT - 9

Illustrate the concept of inter-process communication using shared memory with a C program.

PROGRAM:

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/ipc.h>
#include <sys/shm.h>

#define SHM_SIZE 1024

int main()
{
    key_t key;
    int shmid;
    char *shm_ptr;

    /* Generate a unique key */
    key = ftok("shmfile", 65);
```

```
/* Create shared memory segment */
shmid = shmget(key, SHM_SIZE, IPC_CREAT | 0666);

if(shmid == -1)
{
    perror("shmget");
    exit(EXIT_FAILURE);
}

/* Attach shared memory */
shm_ptr = (char *)shmat(shmid, NULL, 0);

if(shm_ptr == (char *)-1)
{
    perror("shmat");
    exit(EXIT_FAILURE);
}

/* Write data to shared memory */
strcpy(shm_ptr, "Hello, shared memory!");

printf("Data written to shared memory: %s\n", shm_ptr);

/* Detach shared memory */
if(shmdt(shm_ptr) == -1)
{
```

```
        perror("shmdt");
        exit(EXIT_FAILURE);
    }

    /* Remove shared memory segment */
    if(shmctl(shmid, IPC_RMID, NULL) == -1)
    {
        perror("shmctl");
        exit(EXIT_FAILURE);
    }

    return 0;
}
```

OUTPUT:

